Exact Quantum Wave Function Derivation for a Classical Particle in Uniform Gravitational Field

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Abstract

We present a rigorous, first-principles derivation of the stationary spatial wave function $\psi(x, y)$ for a classical body of mass m launched in a uniform gravitational field $g = 10\,\text{m/s}^2$ with initial velocity magnitude u at launch angle θ. By mapping the initial kinematic components to spatial energy parameters and applying the two-dimensional Time-Independent Schrödinger Equation under an impenetrable ground potential boundary $V(y < 0) = \infty$, we decouple the motion into independent horizontal and vertical ordinary differential equations. The horizontal component yields a free plane-wave state characterized by the de Broglie wavenumber k_x, while the vertical component maps directly to Airy's canonical differential equation. Enforcing hard-wall Dirichlet boundary conditions at $y = 0$ and square-integrability at infinity leads to exact normalization constants expressed via Airy function integrals. Finally, we analyze the correspondence limit linking discrete quantum vertical energy states to classical trajectory expectations.

1 Introduction and Physical Model

In classical mechanics, a particle of mass m launched from the origin $(x, y) = (0, 0)$ with initial velocity u at an angle θ relative to the horizontal obeys the deterministic parabolic trajectory governed by Newton's second law:

$$x(t) = (u\cos\theta)t, \quad y(t) = (u\sin\theta)t - \frac{1}{2}gt^2 \tag{1}$$

where $g = 10\,\text{m/s}^2$. To treat this classical body within the paradigm of non-relativistic quantum mechanics, we consider its stationary spatial quantum state governed by the two-dimensional Time-Independent Schrödinger Equation (TISE).

The potential energy landscape $V(x, y)$ consists of a uniform linear gravitational gradient above an impenetrable hard-wall boundary at ground level ($y = 0$):

$$V(x, y) = \begin{cases} mgy, & \text{for } y \geq 0 \\ \infty, & \text{for } y < 0 \end{cases} \tag{2}$$

The domain of spatial existence is $x \in (-\infty, \infty)$ and $y \in [0, \infty)$. The wave function must satisfy Dirichlet boundary conditions:

$$\psi(x, 0) = 0, \quad \lim_{y\to\infty} \psi(x, y) = 0 \tag{3}$$

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2 Separation of Variables

The full 2D Hamiltonian operator $\hat{H}$ for $y \geq 0$ is written as:

$$\hat{H} = -\frac{\overline{h}^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + mgy \tag{4}$$

The corresponding TISE $\hat{H}\psi(x,y) = E\psi(x,y)$ takes the explicit form:

$$-\frac{\overline{h}^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2}\right) + mgy\psi(x,y) = E\psi(x,y) \tag{5}$$

Substituting the separable product ansatz $\psi(x,y) = X(x)Y(y)$ and dividing throughout by $X(x)Y(y)$ isolates the single-variable spatial dependencies:

$$-\frac{\overline{h}^2}{2m}\frac{1}{X(x)}\frac{d^2X(x)}{dx^2} = E - \left(-\frac{\overline{h}^2}{2m}\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2} + mgy\right) \tag{6}$$

Since the left-hand side depends solely on x and the right-hand side depends solely on y, both sides must equal a constant horizontal energy eigenvalue E_x. Defining the vertical energy eigenvalue as $E_y = E - E_x$, we decouple the system into two ordinary differential equations:

$$\frac{d^2X(x)}{dx^2} + k_x^2 X(x) = 0 \tag{7}$$

$$-\frac{\overline{h}^2}{2m}\frac{d^2Y(y)}{dy^2} + mgyY(y) = E_yY(y) \tag{8}$$

where the horizontal wavevector is defined as $k_x \equiv \frac{\sqrt{2mE_x}}{\overline{h}}$.

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3 Horizontal Solution $X(x)$

Classically, the horizontal component of initial kinetic energy is given by:

$$E_x = \frac{1}{2}m(u\cos\theta)^2 \tag{9}$$

Substituting E_x into k_x yields the de Broglie wavenumber along x:

$$k_x = \frac{\sqrt{2m\left(\frac{1}{2}mu^2\cos^2\theta\right)}}{\overline{h}} = \frac{mu\cos\theta}{\overline{h}} \tag{10}$$

The general solution to Eq. (7) for forward motion ($p_x > 0$) is a complex exponential plane wave:

$$X(x) = C_x \exp\left(i\frac{mu\cos\theta}{\overline{h}}x\right) \tag{11}$$

Depending on the spatial boundary choice, C_x takes two canonical representations:

1. **Box Normalization ($x \in [0, L]$):**

$$\int_0^L |X(x)|^2 dx = 1 \implies C_x = \frac{1}{\sqrt{L}} \tag{12}$$

2. **Wavenumber Delta Normalization (k_x-space):**

$$\int_{-\infty}^{\infty} X_{k_x'}^*(x)X_{k_x}(x)dx = \delta(k_x - k_x') \implies C_x = \frac{1}{\sqrt{2\pi}} \tag{13}$$

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4 Vertical Solution $Y(y)$ and Airy Transformation

Rearranging the vertical ODE Eq. (8) highlights the linear spatial potential:

$$-\frac{\overline{h}^2}{2m}\frac{d^2Y(y)}{dy^2} + mg\left(y - \frac{E_y}{mg}\right)Y(y) = 0 \tag{14}$$

We introduce the characteristic gravitational length scale l_g and dimensionless coordinate $z(y)$:

$$l_g = \left(\frac{\overline{h}^2}{2m^2g}\right)^{1/3}, \quad z(y) = \frac{y - \frac{E_y}{mg}}{l_g} = \left(\frac{2m^2g}{\overline{h}^2}\right)^{1/3} y - \left(\frac{2E_y^3}{mg^2\overline{h}^2}\right)^{1/3} \tag{15}$$

Transforming spatial derivatives using the chain rule $\frac{d^2Y}{dy^2} = l_g^{-2}\frac{d^2Y}{dz^2}$ reduces the differential equation to Airy's canonical equation:

$$\frac{d^2Y(z)}{dz^2} - zY(z) = 0 \tag{16}$$

The general solution is a linear combination of linearly independent Airy functions:

$$Y(z) = C_1\mathsf{Ai}(z) + C_2\mathsf{Bi}(z) \tag{17}$$

As $y \to \infty$, $z(y) \to +\infty$. The asymptotic expansions for large positive z are:

$$\mathsf{Ai}(z) \sim \frac{\exp\left(-\frac{2}{3}z^{3/2}\right)}{2\sqrt{\pi}z^{1/4}} \to 0, \quad \mathsf{Bi}(z) \sim \frac{\exp\left(\frac{2}{3}z^{3/2}\right)}{\sqrt{\pi}z^{1/4}} \to \infty \tag{18}$$

Square-integrability mandates setting $C_2 = 0$, yielding the physical vertical wave function:

$$Y(y) = C_y\mathsf{Ai}\left(\left(\frac{2m^2g}{\overline{h}^2}\right)^{1/3} y + z_0\right) \tag{19}$$

where $z_0 \equiv z(0) = -\left(\frac{2E_y^3}{mg^2\overline{h}^2}\right)^{1/3}$.

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5 Hard-Wall Boundary Conditions and Energy Quantization

Enforcing the Dirichlet boundary condition at $y = 0$:

$$Y(0) = 0 \implies \mathsf{Ai}(z_0) = 0 \tag{20}$$

Thus, z_0 must equal one of the discrete negative zeros a_n of the Airy function ($\mathsf{Ai}(a_n) = 0$ for $n = 1, 2, 3, \ldots$):

$$-\left(\frac{2E_{y,n}^3}{mg^2\overline{h}^2}\right)^{1/3} = a_n = -|a_n| \tag{21}$$

Solving for $E_{y,n}$ yields exact, discrete vertical energy eigenvalues:

$$E_{y,n} = \left(\frac{mg^2\overline{h}^2}{2}\right)^{1/3}|a_n| \tag{22}$$

In the classical launch context, the vertical energy expectation corresponds to $E_y = \frac{1}{2}mu^2 \sin^2\theta$. Substituting this parameter directly gives the semi-classical shift argument z_0:

$$z_0 = -\left(\frac{2\left(\frac{1}{2}mu^2\sin^2\theta\right)^3}{mg^2\overline{h}^2}\right)^{1/3} = -\left(\frac{mu^3\sin^3\theta}{2g\overline{h}}\right)^{2/3} \tag{23}$$

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6 Normalization and Synthesis of Wave Function

The vertical normalization factor C_y is determined via spatial integration:

$$\int_0^\infty |Y(y)|^2 dy = 1 \implies C_y^2 l_g \int_{z_0}^\infty \mathrm{Ai}^2(z)dz = 1 \tag{24}$$

Using the indefinite integral identity for Airy functions:

$$\int \mathrm{Ai}^2(z)dz = z\mathrm{Ai}^2(z) - \mathrm{Ai}'^2(z) \tag{25}$$

Evaluating over the domain $[z_0, \infty)$ yields:

$$\int_{z_0}^\infty \mathrm{Ai}^2(z)dz = [0] - \left(z_0\mathrm{Ai}^2(z_0) - \mathrm{Ai}'^2(z_0)\right) = \mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0) \tag{26}$$

Taking the square root gives the complete expression for C_y:

$$C_y = \frac{l_g^{-1/2}}{\sqrt{\mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)}} = \frac{\left(\frac{2m^2g}{\overline{h}^2}\right)^{1/6}}{\sqrt{\mathrm{Ai}'^2(z_0) - z_0\mathrm{Ai}^2(z_0)}} \tag{27}$$

When enforcing discrete boundary quantization ($z_0 = a_n$, $\mathrm{Ai}(a_n) = 0$), C_y simplifies to:

$$C_y = \frac{\left(\frac{2m^2g}{\overline{h}^2}\right)^{1/6}}{|\mathrm{Ai}'(a_n)|} \tag{28}$$

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7 Final Analytical Equations

Combining horizontal and vertical components into $\psi(x, y) = X(x)Y(y)$ provides the complete, normalized wave function solutions.

7.1 General Parameterized Wave Function

In terms of fundamental physical launch parameters $(m, u, \theta, g, \overline{h})$ and horizontal box normalization constant C_x:

$$\psi(x, y) = \mathcal{N} \exp\left(i\frac{mu\cos\theta}{\overline{h}}x\right) \mathrm{Ai}\left(\left(\frac{2m^2g}{\overline{h}^2}\right)^{1/3} y - \left(\frac{mu^3\sin^3\theta}{2g\overline{h}}\right)^{2/3}\right) \tag{29}$$

where the normalization prefactor $\mathcal{N}$ is explicitly given by:

$$\mathcal{N} = \frac{C_x \left(\frac{2m^2 g}{\overline{h}^2}\right)^{1/6}}{\sqrt{\mathsf{Ai}'^2\left(-\left(\frac{mu^3 \sin^3\theta}{2g\overline{h}}\right)^{2/3}\right) + \left(\frac{mu^3 \sin^3\theta}{2g\overline{h}}\right)^{2/3} \mathsf{Ai}^2\left(-\left(\frac{mu^3 \sin^3\theta}{2g\overline{h}}\right)^{2/3}\right)}} \tag{30}$$

7.2 Quantized Boundary Eigenstate Wave Function

For the n-th energy eigenstate where $E_{y,n} = \left(\frac{mg^2\overline{h}^2}{2}\right)^{1/3} |a_n|$:

$$\psi_n(x,y) = \frac{C_x \left(\frac{2m^2 g}{\overline{h}^2}\right)^{1/6}}{|\mathsf{Ai}'(a_n)|} \exp\left(i\frac{mu\cos\theta}{\overline{h}}x\right) \mathsf{Ai}\left(\left(\frac{2m^2 g}{\overline{h}^2}\right)^{1/3} y + a_n\right) \tag{31}$$

8 Conclusion

We have derived the exact quantum mechanical spatial wave function for a classical particle launched in a uniform gravitational field. The spatial wave function transitions from a standard free-particle plane wave horizontally to an Airy function profile vertically. The boundary condition at ground level enforces vertical energy quantization, establishing a continuous spectrum along the horizontal translational axis and a discrete energy spectrum vertically.